Chapter 14

Perturbations of Hydrogen

14.1 Introduction

The nonrelativistic hydrogen Hamiltonian provides one of the central exactly solvable models of quantum mechanics, but it does not describe every feature of the observed hydrogen spectrum. External electric and magnetic fields shift and split the energy levels, while relativistic effects produce smaller corrections even for an isolated atom. These effects are weak (or can be made weak) compared with the Coulomb interaction and therefore provide natural applications of time-independent perturbation theory.

We begin with the Stark effect, which results from placing the atom in an external electric field. We then consider two contributions to the fine structure of hydrogen: the relativistic correction to the electron's kinetic energy and the coupling between its orbital and spin angular momenta. Finally, we will consider the Zeeman effect due to an external magnetic field, distinguishing the regimes in which the magnetic interaction is weak or strong compared with spin-orbit coupling. These perturbations may break symmetries of H_0, lift degeneracies, and select new linear combinations of the unperturbed hydrogen states as the relevant energy eigenstates.

14.2 The Stark Effect

The *Stark effect* refers to the shift and splitting of atomic energy levels caused by an external electric field. We will consider a hydrogen atom placed in a uniform, time-independent electric field

$$\mathbf{E} = \mathcal{E}_0 \hat{\mathbf{z}} \tag{14.1}$$

We take the proton to be fixed at the origin and neglect spin throughout this section. The unperturbed Hamiltonian of the hydrogen atom in the electrostatic model is taken as

$$H_0 = \frac{\mathbf{p}^2}{2m_e} - \frac{k_e e^2}{r} \tag{14.2}$$

with energy eigenstates $|n, \ell, m\rangle$ and corresponding energies

$$E_n^{(0)} = -\frac{13.6 \text{ eV}}{n^2} \tag{14.3}$$

Since the energy depends only on the principal quantum number n, states with different values of ℓ and m may be degenerate.

The uniform electric field in Eq. (14.1) may be derived from the scalar potential

$$\phi(\mathbf{r}) = -\mathbf{E} \cdot \mathbf{r} = -\mathcal{E}_0 z \tag{14.4}$$

and since the electron has charge $-e$, the potential energy is given by

$$V = -e\phi = e\mathcal{E}_0 z \tag{14.5}$$

We will assume the electric field is weak so that V may be treated as a perturbation.

14.2.1 Selection Rules

Before calculating the energy shifts, it's useful to determine which unperturbed states can be connected by the perturbation. We'll find that many of the matrix elements vanish as a result of symmetry. In spherical coordinates,

$$z = r\cos\theta$$

and we notice that the angular dependence of z is proportional to the spherical harmonic $Y_1^0(\theta, \phi)$. In particular,

$$z = r\sqrt{\frac{4\pi}{3}} Y_1^0(\theta, \phi).$$

Consider the matrix element $\langle n', \ell', m'|z|n, l, m\rangle$. Since the hydrogen wave functions separate into radial and angular parts, we can restrict our attention to the angular integral

$$I = \int d\Omega\, (Y_{\ell'}^{m'})^* \cos\theta (Y_\ell^m)$$

The ϕ-dependence of Y_ℓ^m is given by $e^{im\phi}$, so the ϕ-dependent part of the angular integral is

$$\int_0^{2\pi} d\phi\, e^{-im'\phi} e^{im\phi} = \int_0^{2\pi} d\phi\, e^{i(m-m')\phi} = 2\pi\delta_{mm'}.$$

Therefore, the matrix element vanishes unless $m' = m$ or $\Delta m = 0$.

To determine the allowed values of ℓ, we begin by proving the following spherical-harmonic identity:

$$\cos\theta Y_\ell^m = C_{\ell m} Y_{\ell+1}^m + D_{\ell m} Y_{\ell-1}^m$$

where $C_{\ell m}$ and $D_{\ell m}$ are constants that depend on ℓ and m. The θ-dependence of Y_ℓ^m is given by

the associated Legendre polynomial $P_\ell^m(\cos\theta)$, which satisfies the recurrence relation:

$$(\ell - m + 1)\, P_{\ell+1}^m(x) = (2\ell + 1)\, x\, P_\ell^m(x) - (\ell + m)\, P_{\ell-1}^m(x)$$

where $x = \cos\theta$. Rearrange the expression and define constants $A_{\ell m}$ and $B_{\ell m}$ such that

$$x\, P_\ell^m(x) = A_{\ell m} P_{\ell+1}^m(x) + B_{\ell m} P_{\ell-1}^m(x)$$

The spherical-harmonic identity follows directly from the expression above with the appropriate definitions for $C_{\ell m}$ and $D_{\ell m}$. Substituting the identity into the angular integral,

$$\int d\Omega\, (Y_{\ell'}^{m'})^* \cos\theta (Y_\ell^m) = C_{\ell m} \int d\Omega\, (Y_{\ell'}^{m'})^* (Y_{\ell+1}^m) + D_{\ell m} \int d\Omega\, (Y_{\ell'}^{m'})^* (Y_{\ell-1}^m),$$

and using orthonormality,

$$\int d\Omega\, (Y_{\ell'}^{m'})^* \cos\theta (Y_\ell^m) = C_{\ell m} \delta_{\ell',\ell+1} \delta_{m'm} + D_{\ell m} \delta_{\ell',\ell-1} \delta_{m'm}.$$

Consequently, the angular integral can only be nonzero if $m' = m$, which we already knew, and $\ell' = \ell + 1$ or $\ell' = \ell - 1$. Putting the two results together,

$$\big\langle n', \ell', m' \big| z \big| n, l, m \big\rangle \neq 0 \quad \text{iff} \quad \Delta\ell = \pm 1,\ \Delta m = 0 \tag{14.6}$$

It follows that every diagonal matrix element of the perturbation vanishes, but this does not imply that every first-order Stark shift vanishes. If an energy level is degenerate, the off-diagonal matrix elements of V within the degenerate eigenspace must also be considered.

14.2.2 The Ground State

The ground state of hydrogen $|1,0,0\rangle$ is nondegenerate. Its first-order energy correction is

$$E_1^{(1)} = \langle 1,0,0|V|1,0,0\rangle = e\mathcal{E}_0 \langle 1,0,0|z|1,0,0\rangle = 0 \tag{14.7}$$

The second-order energy correction is formally given by

$$E_1^{(2)} = e^2 \mathcal{E}_0^2 \sum_{n' \neq 1} \sum_{\ell', m'} \frac{|\langle n', \ell', m'|z|1,0,0\rangle|^2}{E_1^{(0)} - E_{n'}^{(0)}} \tag{14.8}$$

where the selection rules restrict the contributing intermediate states to those with $\ell' = 1$ and $m' = 0$. Since every excited state has energy greater than the ground state, the second-order correction is negative. Strictly speaking, the complete spectral summation also includes the continuum states, but there's a convenient way to get around that complication. The second-order correction to the energy can be directly calculated from the first-order correction in the state. In Chapter 13, we

used intermediate normalization to show that

$$E = E_n^{(0)} + \lambda \left\langle n^{(0)} \middle| V \middle| \psi \right\rangle, \tag{14.9}$$

in the nondegenerate case. If we substitute the first-order approximation for $|\psi\rangle$ into Eq. (14.9), we obtain an approximation to E that is valid to second order. The second-order correction to the energy can also be written as

$$E_n^{(2)} = \left\langle n^{(0)} \middle| V \middle| n^{(1)} \right\rangle \tag{14.10}$$

where $\left|n^{(1)}\right\rangle$ is the first-order correction to the state. To first-order, we have

$$|\psi\rangle = \left|n^{(0)}\right\rangle + \lambda \left|n^{(1)}\right\rangle + O(\lambda^2), \quad E = E_n^{(0)} + \lambda E_n^{(1)} + O(\lambda^2).$$

Substituting these expressions into the energy eigenvalue equation,

$$(H_0 + \lambda V)\,|\psi\rangle = E\,|\psi\rangle\,,$$

and neglecting terms that are second order and higher, we find

$$H_0 \left|n^{(0)}\right\rangle + \lambda \left(H_0 \left|n^{(1)}\right\rangle + V \left|n^{(0)}\right\rangle\right) = E_n^{(0)} \left|n^{(0)}\right\rangle + \lambda \left(E_n^{(0)} \left|n^{(1)}\right\rangle + E_n(1) \left|n^{(0)}\right\rangle\right).$$

Note that $H_0 \left|n^{(0)}\right\rangle = E_n^{(0)} \left|n^{(0)}\right\rangle$ and $E_n^{(1)} = 0$ for the hydrogen ground state in this perturbation. Therefore, the first-order correction to the energy eigenstate satisfies

$$(H - E_n^{(0)}) \left|n^{(1)}\right\rangle = -V \left|n^{(0)}\right\rangle \tag{14.11}$$

If we can solve Eq. (14.11) to determine the first-order correction, then we can use it to calculate the second-order correction to the energy using Eq. (14.10).

Define natural units (also called atomic units) such that $\hbar = m_e = e = a_0 = 1$, where a_0 is the Bohr radius (in terms of the electron's mass). In these units, the relevant energy and momentum scales are

$$[\text{energy}] = \frac{\hbar^2}{m_e a_0^2}, \quad \text{and} \quad [\text{momentum}] = \frac{\hbar}{a_0},$$

and from the definition of the Bohr radius,

$$a_0 = \frac{\hbar^2}{m_e k_e e^2}$$

it follows that $k_e = 1$. In natural units, the unperturbed Hamiltonian is

$$H_0 = \frac{1}{2}\mathbf{p}^2 - \frac{1}{r}$$

and the perturbation may be written as

$$V = \mathcal{E}_0\, r\cos\theta.$$

where the electric field is measured in units of

$$[\text{electric field}] = \frac{\hbar^2}{m_e a^3}\frac{1}{e} = \frac{k_e e}{a_0^2}.$$

The unperturbed ground-state energy is $E_1^{(0)} = -1/2$. In position-space, Eq. (14.11) becomes

$$\left(-\frac{1}{2}\nabla^2 - \frac{1}{r} + \frac{1}{2}\right)\psi_1^{(1)}(\mathbf{r}) = -\mathcal{E}_0\, r\cos\theta\,\psi_{100}(\mathbf{r}), \tag{14.12}$$

where in natural units, the (unperturbed) ground-state wave function is

$$\psi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi}}e^{-r}.$$

Since the right-hand side is proportional to $\cos\theta$ and ψ_{100}, we introduce the ansatz

$$\psi_1^{(1)}(\mathbf{r}) = -\mathcal{E}_0 f(r)\cos\theta\,\psi_{100}(r) \tag{14.13}$$

for an undetermined function $f(r)$. Substituting the ansatz into Eq. (14.12), and working through some tedious calculations, we arrive at the following differential equation for $f(r)$:

$$-\frac{1}{2}f''(r) + \left(1 - \frac{1}{r}\right)f'(r) + \frac{f(r)}{r^2} = r \tag{14.14}$$

At first it might seem like we've traded one hard problem for another since we have to solve a differential equation, but as it turns out, the solution satisfying the appropriate boundary conditions is given by

$$f(r) = r + \frac{1}{2}r^2, \tag{14.15}$$

as you can easily verify. The radial part of the first-order correction is $R_1^{(1)}(r) \propto f(r)e^{-r}$. Given the solution for $f(r)$ above, $R_1^{(1)} < \infty$ for $r \to 0$ and $R_1^{(1)} \to 0$ for $r \to \infty$. Hence, it is a good solution (see Chapter 9 if you need a refresher on boundary conditions).

Given $\psi_1^{(1)}(\mathbf{r})$, the second-order correction to the ground-state energy can be written as

$$E_1^{(2)} = -\mathcal{E}_0^2\left\langle\left(r^2 + \frac{1}{2}r^3\right)\cos^2\theta\right\rangle_{100}, \tag{14.16}$$

where the angled brackets indicate an expectation value in the unperturbed ground state. The

ground state is spherically symmetric, so the radial and angular parts factorize as

$$E_1^{(2)} = -\mathcal{E}_0^2 \left(\langle r^2 \rangle_{100} + \left\langle \frac{1}{2} r^3 \right\rangle_{100} \right) \langle \cos^2 \theta \rangle_{100} \,, \tag{14.17}$$

For the ground state of hydrogen, if k is a nonnegative integer, then

$$\langle r^k \rangle_{100} = \frac{(k+2)!}{2^{k+1}}, \tag{14.18}$$

and it follows that $\langle r^2 \rangle = 3$ and $\langle r^3 \rangle = 15/2$. Finally, the angular expectation value gives a factor of $1/3$. Altogether, the second-order correction to the energy is

$$E_1^{(2)} = -\frac{9}{4} \mathcal{E}_0^2 \tag{14.19}$$

We convert to ordinary units by replacing

$$E_1^{(2)} \to \frac{E_1^{(2)}}{\hbar^2 / m_e a_0^2}, \quad \text{and} \quad \mathcal{E}_0 \to \frac{\mathcal{E}_0}{k_e e / a_0^2}.$$

Since $k_e = 1/4\pi\varepsilon_0$, we can write the final form of the correction as

$$\boxed{E_1^{(2)} = -9\pi\varepsilon_0 a_0^3 \mathcal{E}_0^2} \tag{14.20}$$

The ground-state energy shift is quadratic in the applied electric field and is therefore called the *quadratic Stark effect*. This energy shift is related to the atomic polarizability of ground-state hydrogen, and it has been measured experimentally. Early experiments used the deflection of atomic hydrogen in an inhomogeneous static electric field, while later measurements used interferometry to determine the polarizability with greater precision.

14.2.3 The First Excited Energy Level

The $n = 2$ energy eigenspace is four-dimensional, with basis

$$|2,0,0\rangle \,, \qquad |2,1,0\rangle \,, \qquad |2,1,1\rangle \,, \qquad |2,1,-1\rangle \,. \tag{14.21}$$

These states all have the same unperturbed energy

$$E_2^{(0)} = -\frac{13.6 \text{ eV}}{4}.$$

Because this energy is degenerate, we must diagonalize the perturbation within the $n = 2$ eigenspace. Let

$$V_{\alpha\beta}^{(2)} = \langle 2, \alpha | V | 2, \beta \rangle \,,$$

where α and β label the four states within the $n = 2$ eigenspace.

The selection rule $\Delta m = 0$ prevents the perturbation from coupling states with different values of m. The selection rule $\Delta\ell = \pm 1$ also prevents V from coupling two $2p$ states to one another. The only nonzero matrix elements within the $n = 2$ eigenspace are therefore

$$\langle 2,0,0|V|2,1,0\rangle \qquad \text{and} \qquad \langle 2,1,0|V|2,0,0\rangle\,.$$

Using the standard hydrogen wave functions,

$$\langle 2,0,0|z|2,1,0\rangle = -3a_0, \tag{14.22}$$

and hence

$$\langle 2,0,0|V|2,1,0\rangle = -3ea_0\mathcal{E}_0. \tag{14.23}$$

In the ordered basis

$$\{|2,0,0\rangle\,, |2,1,0\rangle\,, |2,1,1\rangle\,, |2,1,-1\rangle\}\,,$$

the perturbation matrix is

$$V^{(2)} = 3ea_0\mathcal{E}_0 \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{14.24}$$

The first-order energy corrections are the eigenvalues of this matrix:

$$\boxed{E^{(1)}_{2,+} = +3ea_0\mathcal{E}_0, \qquad E^{(1)}_{2,-} = -3ea_0\mathcal{E}_0, \qquad E^{(1)}_{2,0} = 0,} \tag{14.25}$$

where the zero-order correction remains two-fold degenerate. The corresponding zeroth-order states that evolve smoothly into eigenstates of the perturbed Hamiltonian are

$$|2,+\rangle = \frac{1}{\sqrt{2}}\left(|2,0,0\rangle - |2,1,0\rangle\right), \tag{14.26}$$

$$|2,-\rangle = \frac{1}{\sqrt{2}}\left(|2,0,0\rangle + |2,1,0\rangle\right), \tag{14.27}$$

together with the two unshifted states

$$|2,1,1\rangle\,, \qquad |2,1,-1\rangle\,. \tag{14.28}$$

The states $|2,0,0\rangle$ and $|2,1,0\rangle$ individually have definite parity and therefore have no permanent electric dipole moment along the z-axis. The linear combinations selected by the perturbation do not have definite parity. Indeed,

$$\langle 2,\pm|z|2,\pm\rangle = \pm 3a_0, \tag{14.29}$$

so these states possess equal and opposite electric dipole moments along the direction of the applied

field. The applied electric field lifts the degeneracy of the first excited energy level by selecting states with charge distributions that are polarized either parallel or antiparallel to the field. Because the resulting energy shifts are proportional to the field strength $\mathcal{E}_0$, the splitting is known as the *linear Stark effect*.

The qualitative difference between the $n = 1$ and $n = 2$ Stark effects arises from degeneracy. The nondegenerate ground state has no first-order energy shift and exhibits a quadratic Stark effect. The degenerate $n = 2$ level contains opposite-parity states that can be coupled directly by the perturbation, producing a linear Stark effect.

14.3 Relativistic Corrections

According to the special theory of relativity, the relativistic energy of a particle is given by

$$E = \sqrt{m^2c^4 + p^2c^2}. \tag{14.30}$$

where p is the relativistic momentum. Using a binomial expansion in powers of p/mc,

$$E = mc^2 \left[1 + \frac{1}{2}\left(\frac{p}{mc}\right)^2 - \frac{1}{8}\left(\frac{p}{mc}\right)^4 + \cdots\right]. \tag{14.31}$$

When the relativistic momentum p is sufficiently small compared to mc, we can use the first three terms to approximate the relativistic energy as

$$E \approx mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3c^2}. \tag{14.32}$$

The first term is the rest energy of the particle and the remaining terms constitute an approximation to the kinetic energy. In particular, we can treat the third term as a perturbation to the nonrelativistic kinetic energy $p^2/2m$. Let the perturbation Hamiltonian be

$$V = -\frac{\mathbf{p}^4}{8m_e^3c^2} \tag{14.33}$$

where $\mathbf{p}$ is now the momentum operator and m_e is the mass of the electron.

Although V has arisen as the next term in a low-momentum expansion, we should still verify that it is small compared with the unperturbed hydrogen Hamiltonian. The characteristic momentum scale of an electron in hydrogen is $\hbar/a_0$ and the rest energy of the electron is m_ec^2. We can therefore write the perturbation Hamiltonian as

$$V = -\frac{1}{8}\left(\frac{\hbar/a_0}{m_ec}\right)^4 m_ec^2 \left(\frac{\mathbf{p}}{\hbar/a_0}\right)^4 = -\frac{1}{8}\alpha^4 m_ec^2 \left(\frac{\mathbf{p}}{\hbar/a_0}\right)^4,$$

where $\alpha \approx 1/137$ is the *fine-structure constant*. The operator in parentheses is dimensionless, and its expectation value in a hydrogen energy eigenstate is generally of order one. The characteristic

size of the relativistic correction is therefore $\alpha^4 m_e c^2$. By comparison, the characteristic scale of the nonrelativistic hydrogen binding energy is $\alpha^2 m_e c^2$. Since the relativistic correction is smaller than the nonrelativistic energy by a factor of order α^2, V may be treated as a perturbation.

Now that we've identified the perturbation Hamiltonian, we can apply perturbation theory to determine the resulting energy corrections. The unperturbed hydrogen energies are degenerate with respect to ℓ and m, so we should use degenerate perturbation theory. However, $\mathbf{p}^4$ commutes with both L^2 and L_z, so the perturbation is already diagonal in the basis $|n, \ell, m\rangle$ within each degenerate energy eigenspace. The first-order corrections are simply the diagonal matrix elements of the perturbation. In atomic units, $c = 1/\alpha$, so

$$V = -\frac{\alpha^2}{8}\,\mathbf{p}^4 \tag{14.34}$$

and the first-order relativistic energy correction is

$$E^{(1)}_{\text{rel}} = \langle n, \ell, m|V|n, \ell, m\rangle = -\frac{\alpha^2}{8}\,\langle n, \ell, m|\mathbf{p}^4|n, \ell, m\rangle \tag{14.35}$$

Using

$$\mathbf{p}^2 = 2\left(H_0 + \frac{1}{r}\right),$$

it follows that

$$\begin{aligned}
E^{(1)}_{\text{rel}} &= -\frac{\alpha^2}{8}\,\langle n, \ell, m|\mathbf{p}^2\mathbf{p}^2|n, \ell, m\rangle \\
&= -\frac{\alpha^2}{2}\,\langle n, \ell, m|\left(H_0 + \frac{1}{r}\right)\left(H_0 + \frac{1}{r}\right)|n, \ell, m\rangle \\
&= -\frac{\alpha^2}{2}\,\langle n, \ell, m|\left(E^{(0)}_n + \frac{1}{r}\right)\left(E^{(0)}_n + \frac{1}{r}\right)|n, \ell, m\rangle \\
&= -\frac{\alpha^2}{2}\left[\left(E^{(0)}_n\right)^2 + 2E^{(0)}_n\left\langle\frac{1}{r}\right\rangle_{n\ell m} + \left\langle\frac{1}{r^2}\right\rangle_{n\ell m}\right]
\end{aligned}$$

The two spatial integrals implied in the expression above are much simpler than the ones we would have obtained from the matrix elements of $\mathbf{p}^4$ directly. The radial operator is independent of the angular variables, so the angular parts of the integrals are unity and the matrix elements reduce to radial integrals:

$$\left\langle\frac{1}{r}\right\rangle_{n\ell m} = \int_0^\infty \frac{1}{r}\mathcal{R}^2_{n\ell}(r)r^2 dr = \frac{1}{n^2} \tag{14.36}$$

$$\left\langle\frac{1}{r^2}\right\rangle_{n\ell m} = \int_0^\infty \frac{1}{r^2}\mathcal{R}^2_{n\ell}(r)r^2 dr = \frac{1}{n^3\left(\ell + \frac{1}{2}\right)} \tag{14.37}$$

Using $E_n^{(0)} = -1/2n^2$, the first-order relativistic correction in atomic units is

$$E_{\text{rel}}^{(1)} = -\frac{1}{2}\alpha^2 \left[\frac{1}{n^3\left(\ell + \frac{1}{2}\right)} - \frac{3}{4n^4} \right] \tag{14.38}$$

To convert the expression to ordinary units, multiply by the energy scale

$$\frac{\hbar^2}{m_e a_0^2} = \alpha^2 m_e c^2.$$

Hence,

$$\boxed{E_{\text{rel}}^{(1)} = -\frac{1}{2}\alpha^4 m_e c^2 \left[\frac{1}{n^3\left(\ell + \frac{1}{2}\right)} - \frac{3}{4n^4} \right]} \tag{14.39}$$

As we expected, the relativistic correction is α^2 times smaller than the Bohr energy.

State	$E_{\text{rel}}^{(1)}$ $(10^{-4}\,\text{eV})$
$1s$	-9.06
$2s$	-1.47
$2p$	-0.264

Table 14.1: First-order relativistic corrections.

14.4 Spin-Orbit Coupling

The electron's orbital motion and intrinsic spin are accompanied by magnetic moments. In the electron's rest frame, the electric field of the proton is partly experienced as a magnetic field, which interacts with the electron's spin magnetic moment. After accounting for the relativistic Thomas-precession correction, this interaction is described by the spin-orbit Hamiltonian

$$V_{\text{SO}} = \frac{k_e e^2}{2m_e c^2 r^3} \mathbf{L} \cdot \mathbf{S} \tag{14.40}$$

In atomic units, where $\hbar = m_e = a_0 = e = 1$ and $\alpha = 1/c$,

$$V_{\text{SO}} = \frac{\alpha^2}{2r^3} \mathbf{L} \cdot \mathbf{S} \tag{14.41}$$

For low-energy states with small angular-momentum quantum numbers,

$$r \approx a_0, \quad L \approx S \approx \hbar,$$

so the characteristic size of the spin-orbit interaction is $V_{\text{SO}} \approx \alpha^2$ in atomic units, or $\alpha^4 m_e c^2$ in ordinary units (recall the energy unit may be written as $\alpha^2 m_e c^2$). Evidently, the spin-orbit interaction is on the same order of magnitude as the relativistic correction we studied in the

previous section.

Including the electron's spin degrees of freedom, the eigenstates of the unperturbed Hamiltonian in the uncoupled basis are

$$|n,\ell,m\rangle\,|s,m_s\rangle = |n,\ell,m;m_s\rangle$$

where $|n,\ell,m\rangle$ denotes the electron's orbital state and $|s,m_s\rangle$ describes its spin state. The operator $\mathbf{L}\cdot\mathbf{S}$ is not diagonal in the uncoupled basis, but it is diagonal in the *coupled basis*, defined as the simultaneous eigenstates of L^2, S^2, J^2 and J_z, where

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \tag{14.42}$$

is the total angular momentum. Denote the elements of the coupled basis as $|n,\ell,j,m_j\rangle$. Since the electron has $s = 1/2$, the possible values of j are

$$j = \begin{cases} \ell \pm \frac{1}{2}, & \ell > 0, \\ \\ \frac{1}{2}, & \ell = 0. \end{cases} \tag{14.43}$$

Using

$$J^2 = (\mathbf{L}+\mathbf{S})\cdot(\mathbf{J}+\mathbf{S}) = L^2 + S^2 + 2\mathbf{L}\cdot\mathbf{S},$$

we obtain

$$\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}\left(J^2 - L^2 - S^2\right),$$

and furthermore, since $s = 1/2$,

$$\mathbf{L}\cdot\mathbf{S}\,|n,\ell,j,m_j\rangle = \frac{1}{2}\left[j(j+1) - \ell(\ell+1) - \frac{3}{4}\right]|n,\ell,j,m_j\rangle\,. \tag{14.44}$$

Using time-independent perturbation theory, the first-order correction to the energy is

$$E^{(1)}_{\rm SO} = \langle n,\ell,j,m_j|V_{\rm SO}|n,\ell,j,m_j\rangle \tag{14.45}$$

The radial and angular factors separate, and thus

$$E^{(1)}_{\rm SO} = \frac{\alpha^2}{2}\left\langle\frac{1}{r^3}\right\rangle_{n\ell}\langle\mathbf{L}\cdot\mathbf{S}\rangle_{\ell j} \tag{14.46}$$

For $\ell \neq 0$,

$$\left\langle\frac{1}{r^3}\right\rangle_{n\ell} = \frac{1}{n^3\ell(\ell+1/2)(\ell+1)}, \tag{14.47}$$

and we conclude that

$$E^{(1)}_{\rm SO} = \frac{\alpha^2}{4n^3}\,\frac{j(j+1)-\ell(\ell+1)-3/4}{\ell(\ell+1/2)(\ell+1)}. \tag{14.48}$$

For $j = \ell + 1/2$, the orbital and spin angular momenta are more nearly aligned, so $\langle\mathbf{L}\cdot\mathbf{S}\rangle > 0$,

and the spin-orbit interaction raises the energy. For $j = \ell - 1/2$, the angular momenta are more nearly oppositely aligned, so $\langle \mathbf{L} \cdot \mathbf{S} \rangle < 0$, and the spin-orbit interaction lowers the energy. The correction does not depend on m_j, so the degeneracy among states with different values of m_j remains. Spin-orbit coupling instead lifts the degeneracy between states with the same n and ℓ but different values of j. For $\ell = 0$, $\mathbf{L} \cdot \mathbf{S} = 0$, so there simply is no spin-orbit correction. A complete relativistic treatment of the s-states also contains the *Darwin term*.

Restoring ordinary units, the first-order energy correction is

$$E_{\text{SO}}^{(1)} = \frac{\alpha^4 m_e c^2}{4n^3} \, \frac{j(j+1) - \ell(\ell+1) - 3/4}{\ell(\ell+1/2)(\ell+1)}, \tag{14.49}$$

which is on the order of $\alpha^4 m_e c^2$ for states with low angular-momentum quantum numbers. Adding the spin-orbit correction to the relativistic correction gives the complete fine-structure formula

$$E_{\text{fs}}^{(1)} = -\frac{\alpha^4 m_e c^2}{2n^4} \left[\frac{n}{j+1/2} - \frac{3}{4} \right]. \tag{14.50}$$

Since the unperturbed hydrogen energies are

$$E_n^{(0)} = -\frac{\alpha^2 m_e c^2}{2n^2} = -\frac{13.6 \text{ eV}}{n^2}, \tag{14.51}$$

the energy levels, including the leading fine-structure correction, are

$$\boxed{E_{nj} = -\frac{13.6 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right) \right].} \tag{14.52}$$

The correction depends on n and j, but not separately on ℓ or m_j. Thus, it follows from our corrected model that states with the same n and j remain degenerate.

14.5 Zeeman Effect

When an atom is placed in an external magnetic field, its energy levels generally split into several closely spaced levels. This phenomenon is called the Zeeman effect. The splitting arises because the magnetic moments associated with the atom's orbital and spin angular momenta interact with the applied magnetic field. The Zeeman effect applies to all atoms, but we will focus primarily on hydrogen.

14.5.1 Zeeman Effect without Spin

We begin by ignoring the electron's intrinsic spin. Even though the electron's orbital and spin magnetic moments contribute equally to the effect, we will use this unrealistic model because it is an easy calculation and it introduces us to the essential features of the Zeeman effect. Because we are ignoring spin, the eigenstates of the hydrogen Hamiltonian are $|n\ell m_\ell\rangle$ and the energies are

$E_n^{(0)} = -13.6 \text{ eV}/n^2$. We perturb the system by applying a magnetic field $\mathbf{B}$ along the z axis. The resultant Zeeman perturbation Hamiltonian is

$$V_{\text{Z}} = g_\ell \mu_B \frac{1}{\hbar} \mathbf{L} \cdot \mathbf{B} = g_\ell \mu_B \frac{B}{\hbar} L_z, \tag{14.53}$$

where μ_B is the Bohr magneton and $g_\ell = 1$ is the electron's orbital g-factor. The magnetic moment of the proton is about 1000 times smaller than that of the electron, so we will neglect the proton contribution to the Zeeman effect.

Since the zeroth-order energy eigenstates are eigenstates of the perturbation, the first-order energy corrections are the diagonal elements of the perturbation Hamiltonian:

$$\boxed{E_n^{(1)} = \langle n\ell m_\ell | V_{\text{Z}} | n\ell m_\ell \rangle = g_\ell \mu_B B m_\ell.} \tag{14.54}$$

For a given value of ℓ, the magnetic quantum number takes the values $m_\ell = -\ell, -\ell+1, \ldots, \ell$, so an orbital level splits into $2\ell + 1$ equally spaced levels. The separation between adjacent levels is $\Delta E = \mu_B B$. This result is called the normal Zeeman effect. It correctly describes systems in which spin plays no role, but it is not sufficient for a single electron such as the electron in hydrogen. The normal Zeeman effect is proportional to the orbital g-factor, the Bohr magneton μ_B, the magnetic field strength B, and the magnetic quantum number m_ℓ. This general form is repeated when we study the Zeeman effect in more realistic models. The g-factor and the particular magnetic quantum number change as we include other magnetic moments in the model.

14.5.2 Including Electron Spin

Including spin, the full Zeeman interaction is given by the Hamiltonian

$$V_{\text{Z}} = \frac{\mu_B B}{\hbar} \Big(g_\ell L_z + g_e S_z \Big) \tag{14.55}$$

Taking $g_\ell = 1$ and $g_e = 2$, one might be tempted to conclude that the Zeeman shift is

$$\Delta E_{\text{Z}} = \mu_B B (m_\ell + 2 m_s).$$

but this is only correct when the energy eigenstates can be labeled by the quantum numbers m_ℓ and m_s. Recall that the spin-orbit interaction is diagonal in the coupled basis while the Zeeman interaction is diagonal in the uncoupled basis. Which basis we choose for the zeroth-order energy eigenstates depends on the relative strength of the interactions.

14.5.3 Strong Magnetic Field

In the strong-field regime, the Zeeman interaction is strong enough to disrupt the coupling between $\mathbf{L}$ and $\mathbf{S}$. We include the Zeeeman Hamiltonian in zeroth order and treat the fine structure as a perturbation. The uncoupled basis, denoted $|n\ell m_\ell m_s\rangle$, is the preferred basis because the Zeeman

Hamiltonian is diagonal in that basis. The zeroth-order energies are

$$\begin{aligned} E_n^{(0)} &= -\frac{13.6\text{ eV}}{n^2} + \langle n\ell m_\ell m_s | V_{\text{Z}} | n\ell m_\ell m_s \rangle \\ &= -\frac{13.6\text{ eV}}{n^2} + \mu_B B(m_\ell + 2m_s) \end{aligned} \tag{14.56}$$

Now we treat the fine structure as a perturbation to the zeroth-order states that include the Zeeman interaction. The relativistic Hamiltonian is diagonal in the both the coupled and uncoupled bases, but the spin-orbit Hamiltonian is diagonal only in the coupled basis. Fortunately, the off-diagonal matrix elements of the spin-orbit Hamiltonian do not couple any of the states that are degenerate with respect to the Zeeman Hamiltonian. Hence we can use nondegenerate perturbation theory to find the first-order fine-structure corrections. In the uncoupled basis, the spin-orbit energy corrections are the expectation values

$$\begin{aligned} E_{\text{SO}}^{(1)} &= \frac{k_e e^2}{2m_e^2 c^2} \langle m\ell m_\ell m_s | \left(\frac{1}{r^3} \mathbf{L} \cdot \mathbf{S} \right) | m\ell m_\ell m_s \rangle \\ &= \frac{k_e e^2}{2m_e^2 c^2} \left\langle \frac{1}{r^3} \right\rangle_{n\ell} \langle m_\ell m_s | \tfrac{1}{2} \left(L_+ S_- + L_- S_+ \right) + L_z S_z | m_\ell m_s \rangle \end{aligned} \tag{14.57}$$

Because the ladder operators connect adjacent states, they produce no diagonal matrix elements, so the first-order correction reduces to

$$E_{\text{SO}}^{(1)} = \frac{k_e e^2}{2m_e^2 c^2} \left\langle \frac{1}{r^3} \right\rangle_{n\ell} m_\ell m_s \hbar^2 \tag{14.58}$$

Using Eq. (14.47) for the radial matrix elements (in ordinary units), we obtain

$$E_{\text{SO}}^{(1)} = \frac{1}{2} \alpha^4 m_e c^2 \frac{m_\ell m_s}{n^3 \ell(\ell + 1/2)(\ell + 1)}. \tag{14.59}$$

Adding together the relativistic and spin-orbit corrections yields the fine-structure shifts

$$\boxed{E_{\text{fs}}^{(1)} = \frac{1}{2} \alpha^4 m_e c^2 \left[\frac{3}{4n^3} - \frac{\ell(\ell + 1) - m_\ell m_s}{n^3 \ell(\ell + 1/2)(\ell + 1)}. \right]} \tag{14.60}$$

14.5.4 Weak Magnetic Field

In the weak-field regime, the zeroth-order states are the coupled basis states $|n\ell j m_j\rangle$, in which the fine-structure Hamiltonian is diagonal. Our task is to determine how the Zeeman perturbation modifies the corresponding zeroth-order energies. The fine-structure energy depends on j, but not on m_j. Thus, for fixed n, ℓ, and j, the $2j + 1$ states with different values of m_j all have the same unperturbed energy. Each j-level therefore contains a $(2j + 1)$-dimensional degenerate subspace.

We must consequently use degenerate perturbation theory, which ordinarily requires us to di-

agonalize the perturbation within each degenerate subspace. In this case, however, the coupled states $|n\ell j m_j\rangle$ already form a basis in which $V_{\rm Z}$ is diagonal within each fixed-j subspace. Although the Zeeman perturbation is not diagonal in the entire coupled basis (it can couple states with the same m_j but different j), perturbation theory requires only that it be diagonalized within each degenerate subspace. The different j-levels are separated by the fine-structure splitting and are therefore nondegenerate in the weak-field treatment.

The simplest way to understand this result is to consider the commutator of $V_{\rm Z}$ with J_z. Since J_z commutes with L_z and S_z, it follows that $[V_{\rm Z}, J_z] = 0$, and furthermore, the states $|n\ell j m_j\rangle$ must be the "good states" to use for perturbation theory (see Chapter #13). Hence,

$$E_{\rm Z}^{(1)} = \langle n\ell j m_j | V_{\rm Z} | n\ell j m_j \rangle \,, \tag{14.61}$$

and all that remains is to compute the expectation value. There is an elegant way to calculate the first-order correction that comes from the Wigner-Eckhart theorem. The only formula we need from the Wigner-Eckhart theorem relates the matrix element of any general vector component A_z to the matrix element of the total angular momentum component J_z:

$$\big\langle j m_j \big| A_z \big| j m_j' \big\rangle = \frac{\langle j m_j | \mathbf{A}\cdot\mathbf{J} | j m_j \rangle}{\hbar^2 j(j+1)} \big\langle j m_j \big| J_z \big| j m_j' \big\rangle \tag{14.62}$$

This is known as the projection theorem, and there is a heuristic way of understanding it using a classical model. The classical picture is that, within a state of definite total angular momentum j, the only distinguished direction is the direction of $\mathbf{J}$. Therefore, the part of any vector $\mathbf{A}$ that can survive averaging is its projection along $\mathbf{J}$. Classically, decompose

$$\mathbf{A} = \mathbf{A}_\parallel + \mathbf{A}_\perp, \quad \text{where} \quad \mathbf{A}_\parallel = \frac{\mathbf{A}\cdot\mathbf{J}}{J^2}\,\mathbf{J}.$$

If $\mathbf{A}$ precesses about $\mathbf{J}$, the perpendicular component continually changes direction and averages to zero, i.e., $\overline{\mathbf{A}}_\perp = 0$. Thus,

$$\overline{\mathbf{A}} = \frac{\mathbf{A}\cdot\mathbf{J}}{J^2}\mathbf{J}, \quad \text{and} \quad \overline{A}_z = \frac{\mathbf{A}\cdot\mathbf{J}}{J^2} J_z.$$

Because J_z is diagonal in the coupled basis, the only nonzero matrix elements have $m_j = m_j'$, and the projection theorem reduces to

$$\langle j m_j | A_z | j m_j \rangle = \frac{m_j}{\hbar j(j+1)} \langle j m_j | \mathbf{A}\cdot\mathbf{J} | j m_j \rangle \tag{14.63}$$

Using

$$\mathbf{L}\cdot\mathbf{J} = \tfrac{1}{2}\left(\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2\right), \quad \text{and} \quad \mathbf{S}\cdot\mathbf{J} = \tfrac{1}{2}\left(\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2\right),$$

the diagonal matrix elements required in the projection theorem are

$$\langle \mathbf{L} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} \left[j(j+1) + \ell(\ell+1) - s(s+1) \right] \tag{14.64}$$

$$\langle \mathbf{S} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} \left[j(j+1) + s(s+1) - \ell(\ell+1) \right] \tag{14.65}$$

Using these coefficients in the projection theorem yields the Lz and Sz matrix elements in the coupled basis within a given j subspace:

$$\langle L_z \rangle = \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2j(j+1)} \, m_j \tag{14.66}$$

$$\langle S_z \rangle = \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \, m_j \tag{14.67}$$

The resulting first-order Zeeman correction to the energy can be written in the standard Zeeman form as

$$E_{\mathrm{Z}}^{(1)} = g_j \mu_B B m_j, \tag{14.68}$$

where g_j is the Landé g-factor given by

$$g_j = g_\ell \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2j(j+1)} + g_e \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \tag{14.69}$$

The Landé g-factor accounts for the relative contributions of the magnetic moments due to the spin and orbital angular momenta caused by their differing magnitudes and differing alignments. For hydrogen, with only one electron, $s = 1/2$ and $j = \ell \pm 1/2$. Using $g_\ell = 1$ and $g_e = 2$, the Landé g-factor for hydrogen is

$$g_j = 1 \pm \frac{1}{2\ell + 1}. \tag{14.70}$$

Consequently, the Zeeman energy correction is dependent on j and ℓ:

$$\boxed{E_{\mathrm{Z}}^{(1)} = \mu_B B m_j \left(1 \pm \frac{1}{2\ell + 1} \right).} \tag{14.71}$$

14.6 Summary: Choosing the Right States

For a nondegenerate unperturbed state $|n\rangle$, the first-order energy correction is simply

$$E_n^{(1)} = \langle n|V|n\rangle \, .$$

For a degenerate energy level, however, an arbitrary unperturbed basis state does not necessarily evolve into an energy eigenstate when the perturbation is turned on. One must first diagonalize the perturbation within the degenerate eigenspace. Once these "good states" have been identified,

their first-order energy corrections are again given by expectation values:

$$E_n^{(1)} = \langle n|V|n\rangle\,.$$

Thus, throughout this chapter, the important question was whether the states being used already diagonalized the perturbation *within each degenerate subspace*.

14.6.1 Stark Effect

The hydrogen ground state $|1,0,0\rangle$ is nondegenerate, so ordinary nondegenerate perturbation theory applies:

$$E_1^{(1)} = \langle 1,0,0|eE_0 z|1,0,0\rangle = 0.$$

The expectation value vanishes by parity, so the leading ground-state Stark shift is second order. The $n=2$ level is degenerate, and the states $|2,0,0\rangle$ and $|2,1,0\rangle$ are coupled by the perturbation. Their individual expectation values vanish, but they are not the appropriate states for first-order perturbation theory. Diagonalizing V within the $n=2$ eigenspace gives

$$\frac{1}{\sqrt{2}}\left(|2,0,0\rangle \pm |2,1,0\rangle\right),$$

together with the uncoupled states $|2,1,\pm1\rangle$. These states diagonalize the perturbation within the degenerate subspace, so their first-order shifts may be calculated as expectation values:

$$E_{2,\pm}^{(1)} = \langle 2,\pm|V|2,\pm\rangle = \pm 3ea_0E_0, \qquad E_{2,1,\pm1}^{(1)} = 0.$$

14.6.2 Relativistic Correction

The Coulomb energies are degenerate with respect to ℓ and m, so degenerate perturbation theory is formally required. Nevertheless, the relativistic perturbation

$$V_{\text{rel}} = -\frac{\mathbf{p}^4}{8m_e^3c^2}$$

is rotationally invariant and commutes with L^2 and L_z. It therefore does not mix states with different ℓ or m within a fixed-n eigenspace. The states $|n,\ell,m\rangle$ already diagonalize the perturbation within every degenerate energy eigenspace, and hence

$$E_{\text{rel}}^{(1)} = \langle n,\ell,m|V_{\text{rel}}|n,\ell,m\rangle\,.$$

14.6.3 Spin-Orbit Coupling

The spin-orbit perturbation is proportional to

$$V_{\text{SO}} \propto \frac{1}{r^3}\mathbf{L}\cdot\mathbf{S}.$$

It is not diagonal in the uncoupled basis $|n, \ell, m_\ell, m_s\rangle$, because the ladder-operator terms in $\mathbf{L} \cdot \mathbf{S}$ mix different values of m_ℓ and m_s. It is diagonal in the coupled basis $|n, \ell, j, m_j\rangle$, since

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}\left(J^2 - L^2 - S^2\right).$$

The coupled states are therefore the appropriate zeroth-order states, and

$$E_{\mathrm{SO}}^{(1)} = \langle n, \ell, j, m_j | V_{\mathrm{SO}} | n, \ell, j, m_j \rangle .$$

The result depends on n, ℓ, and j, but not on m_j, so the degeneracy among the different m_j states is not lifted.

14.6.4 Strong-Field Zeeman Effect

In the strong-field regime, the Zeeman interaction is included in the zeroth-order Hamiltonian because it is stronger than spin-orbit coupling. The Zeeman Hamiltonian is diagonal in the uncoupled basis $|n, \ell, m_\ell, m_s\rangle$, so these are the appropriate zeroth-order states. The fine-structure Hamiltonian is then treated as the perturbation. The spin-orbit operator is not completely diagonal in the uncoupled basis, but its off-diagonal ladder-operator terms do not connect states that have the same zeroth-order Zeeman energy. Therefore, within every degenerate zeroth-order subspace, the fine-structure perturbation is already diagonal in the relevant uncoupled states. See the soultions to Homework #4, Problem #3. Its first-order correction can then be calculated using

$$E_{\mathrm{fs}}^{(1)} = \langle n, \ell, m_\ell, m_s | V_{\mathrm{fs}} | n, \ell, m_\ell, m_s \rangle .$$

In particular, only the diagonal term $L_z S_z$ contributes to the first-order expectation value of the spin-orbit interaction.

14.6.5 Weak-Field Zeeman Effect

In the weak-field regime, the fine structure is included in the zeroth-order Hamiltonian, so the appropriate states are the coupled states $|n, \ell, j, m_j\rangle$. States with different m_j remain degenerate before the magnetic field is applied. We must therefore determine whether the Zeeman perturbation is diagonal within each degenerate subspace. Because

$$[V_{\mathrm{Z}}, J_z] = 0,$$

the Zeeman perturbation cannot connect states with different m_j. Within a fixed n, ℓ, and j subspace, each value of m_j labels a distinct state, so the coupled basis diagonalizes V_{Z} within that subspace. Moreover, V_{Z} does not mix different values of ℓ, while states with different j are separated

by the fine-structure splitting and are not degenerate in the weak-field treatment. Therefore,

$$E_Z^{(1)} = \langle n, \ell, j, m_j | V_Z | n, \ell, j, m_j \rangle = g_j \mu_B B m_j.$$

The strong- and weak-field limits require different states because different interactions define the zeroth-order Hamiltonian:

Regime	Dominant interaction	Appropriate states
Strong magnetic field	V_{Z}	$\lvert n, \ell, m_\ell, m_s \rangle$
Weak magnetic field	V_{fs}	$\lvert n, \ell, j, m_j \rangle$

In each case, the appropriate states are those that diagonalize the zeroth-order Hamiltonian and the perturbation within any remaining degenerate subspaces.